

Project Review and Verification Log

Math 464 Lunar Power Allocation Capstone

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February 7, 2026 – April 28, 2026

Review Method and Scope

This document records the AI-assisted review process used during development of the Math 464 lunar power-allocation capstone report and accompanying Python implementation. The review was conducted with the OpenAI 5.2 model. The purpose of these exchanges was to audit the project for internal consistency, linear-programming formulation correctness, source/data alignment, numerical validity, implementation consistency, and whether report claims were supported by the Python output.

The entries below are organized by review date. Each entry includes the audit prompt and the model's corresponding review notes.

Review date: February 7, 2026

Audit request for: Final_Project_draft1(2).pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Abstract, Introduction, Mission scenario, Research context, Data/assumptions, Source table, LP formulation, Standard form, Implementation, Python script, Verification examples, Results, Runtime comparison, Figures, Appendix, Sources/acknowledgment

C) ISSUE NOTES (NO ADVICE):

Abstract: Contains placeholder/final-result language and claims a complete three-method comparison before the script supports it.

Abstract: Reports objective values and shortage patterns that are not traceable to shown output.

Introduction: Project scope is described generally as lunar power allocation, but the specific optimization question is not yet stated.

Introduction: The role of Part 1 implementation is disconnected from the Part 2 lunar LP.

Introduction: Includes draft notes/placeholders rather than finished exposition.

Mission scenario: "Foundation habitat" phase is named but not defined or source-backed.

Mission scenario: Crewed duration, dormant duration, crew size, and source architecture are missing or inconsistently stated.

Research context: NASA/source references are incomplete, and some claims are not tied to citations.

Data/assumptions: Source-side power values, load values, efficiencies, and penalty weights are listed without separating source-backed data from modeling assumptions.

Data/assumptions: Total available power and flexible available power are mixed.

Data/assumptions: The fixed 2 kW habitat keep-alive load is not handled consistently.

Source table: Several entries are blank, placeholder, or marked “source” without a real source/status explanation.

LP formulation: Decision variables are incomplete.

LP formulation: Dispatch variables, delivered-power variables, and shortage variables are not all defined.

LP formulation: Demand constraints are incomplete or only described in prose.

LP formulation: Delivery-efficiency equations are missing or inconsistent.

LP formulation: Capacity constraints are not written in a complete auditable form.

LP formulation: Objective function is incomplete and does not consistently include operating-hour multipliers.

Standard form: No complete simplex-ready $Az = b, z \geq 0$ form is given.

Implementation: Tableau simplex and revised simplex are claimed but not actually described.

Implementation: General-form conversion and initial BFS logic are missing.

Python script: Uses PuLP as the primary solve method, so the custom-simplex requirement is not satisfied.

Python script: The script has incomplete functions/placeholders and cannot be verified as a working final script.

Verification examples: No diagnostic LPs are shown for conversion, Phase I, infeasibility, or mixed constraint senses.

Results: Reported objective values are unsupported by current script output.

Runtime comparison: Runtime tables/plots are missing or placeholders.

Figures: Figure references are present before the figures/results are consistent.

Appendix: Appendix output is missing.

Sources/acknowledgment: Source list and tool-use acknowledgment are incomplete.

D) STOP

Audit request for: Final_Project_draft2(2).pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Abstract, Introduction, Mission scenario, Research context, Data/assumptions, Source table, LP formulation, Standard form, Implementation, Python script, Verification examples, Results, Runtime comparison, Figures, Appendix

C) ISSUE NOTES (NO ADVICE):

Abstract: Still claims final agreement between PuLP, tableau simplex, and revised simplex before all three methods are verified.

Abstract: Large-case objective and shortage claims remain unsupported by displayed computations.

Introduction: The application is clearer, but the connection between diagnostic simplex testing and the lunar benchmark is still incomplete.

Mission scenario: Crewed and dormant modes are now mentioned, but the 30-day and 335-day durations are not consistently used.

Mission scenario: Foundation Surface Habitat / Foundational Exploration terminology is still not clearly explained.

Research context: Some source claims remain uncited or too broad.

Data/assumptions: Source-backed values and adopted modeling choices are still blended together.

Data/assumptions: The draft still mixes 50, 40 total power with 48, 38 flexible power without a clean preprocessing explanation.

Data/assumptions: Remote efficiencies are listed as if they are sourced physical constants rather than assumed model values.

Source table: Some rows still contain vague labels such as “NASA” or “assumed” without specifying which values are adopted.

LP formulation: Variables are now partly defined, but $y_{i,m}$, $x_{i,m}$, and $s_{i,m}$ are not consistently used afterward.

LP formulation: The hard habitat keep-alive load is still sometimes treated as flexible.

LP formulation: Capacity and demand constraints are incomplete for all modes/loads.

LP formulation: Objective still omits or inconsistently applies mode-hour multipliers.

Standard form: Slack/artificial-variable construction is not fully shown.

Implementation: General-form simplex conversion is claimed but not demonstrated.

Implementation: Phase I or Big-M initialization is still missing/incomplete.

Python script: Tableau simplex is not clearly separated from the solver baseline.

Python script: Revised simplex is named but does not yet appear to be an independent implementation.

Verification examples: No shown tests verify $\leq$, $\geq$, equality, and infeasible cases.

Results: Small/medium/large dataset sizes do not match the model dimensions used in the text.

Runtime comparison: Runtime figure/table is missing or based on placeholder values.

Appendix: Verification output is still absent.

D) STOP

Audit request for: Final_Project_draft3.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not.

Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Abstract, Mission scenario, Data/assumptions, Source table, LP formulation, Standard form, Implementation, Python script, Verification examples, Results, Runtime comparison, Figures

C) ISSUE NOTES (NO ADVICE):

Abstract: Still reports final numerical conclusions before they are fully supported by reproducible output.

Mission scenario: Crewed/dormant structure is improved, but source-backed motivation is still compressed and partly unsupported.

Data/assumptions: The paper states $P_c = 50$ and $P_d = 40$, while the script uses 48 and 38 without clearly explaining fixed-load subtraction.

Data/assumptions: Habitat keep-alive preprocessing remains unclear.

Data/assumptions: Some demand values are still written as if directly source-backed rather than adopted representative values.

Source table: Source-status distinction is incomplete for ISRU split, remote efficiencies, and penalty weights.

LP formulation: Delivered-power and source-side dispatch variables are both used, but $x_{i,m} = \eta_i y_{i,m}$ is not enforced consistently.

LP formulation: Demand equations are missing for some loads/modes.

LP formulation: Capacity constraints use total source power in one place and flexible capacity in another.

LP formulation: Objective uses weighted shortage, but units are unclear because hours are inconsistent.

Standard form: Variable counts in the report do not match the standard-form construction.

Implementation: General-form solver pipeline is described more broadly than the code supports.

Implementation: Reduced-cost sign convention for minimization is not stated.

Python script: Phase I logic is incomplete, so infeasibility detection is not supported as written.

Python script: Some outputs still appear copied/manually summarized rather than printed by the current script.

Verification examples: Diagnostic examples are referenced but not included with expected outputs.

Results: Large-case objective is reported without enough allocation/shortage detail to verify it.

Runtime comparison: The comparison does not track both decision-variable count and constraint count.

Figures: Allocation figure is referenced but not clearly tied to delivered versus dispatched power.

D) STOP

Review date: February 14, 2026

Audit request for: Final_Project_draft4.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not.

Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Data/assumptions, Source table, LP formulation, Standard form, Implementation, Python script, Verification examples, Results, Runtime comparison

C) ISSUE NOTES (NO ADVICE):

Data/assumptions: Total source power versus flexible available power remains unclear.

Data/assumptions: The fixed habitat keep-alive load is sometimes subtracted before optimization and sometimes included as a flexible demand.

Data/assumptions: Remote efficiencies 0.95, 0.92, and 0.90 are still presented too strongly as source-derived.

Source table: The source/modeling-status column is improved but still incomplete for several adopted values.

LP formulation: Dispatch, delivered, and shortage variables are now listed but not consistently connected across the equations.

LP formulation: The hard habitat requirement is stated but not clearly separated from the flexible-load LP.

LP formulation: Demand balance $x_{i,m} + s_{i,m} = d_{i,m}$ is missing or inconsistent in one section.

LP formulation: Objective includes hours in one location but not in another.

Standard form: Slack variables are discussed, but the full $Az = b, z \geq 0$ form is not auditable yet.

Implementation: Tableau and revised simplex are described generically, but the actual basis update/pivot convention is unclear.

Python script: The lunar matrix builder and PuLP builder appear to use different model structures without explanation.

Python script: Revised simplex still appears dependent on tableau results rather than independently solving.

Python script: Phase I artificial variables are not clearly removed from the final problem.

Verification examples: No shown example verifies that Phase I finds an initial BFS.

Results: The report says all three methods agree, but the script output does not yet show all three methods side-by-side.

Runtime comparison: Some runtimes appear copied from earlier runs or not generated by the current script.

D) STOP

Audit request for: Final_Project_draft5.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Data table, LP formulation, Standard form, Implementation, Python script, Verification examples, Results, Figures

C) ISSUE NOTES (NO ADVICE):

Data table: Some values are still called NASA source-backed even though they are modeling splits or representative assumptions.

Data table: Dormant ISRU split and remote efficiencies are not clearly marked as assumptions.

LP formulation: Habitat keep-alive is described as hard-required, but the model still treats it like a flexible load in places.

LP formulation: Delivered-power variables and dispatch variables still do not match the matrix used

in the code.

LP formulation: The objective is mostly correct, but one displayed equation still omits operating-hour multipliers.

Standard form: The formula for standard-form dimensions does not match the code's variable construction.

Implementation: Bland's rule is named, but the entering/leaving rule is not fully stated.

Implementation: Phase I explanation is incomplete.

Python script: The available-power dictionary is named as if it contains total source power but actually contains post-keep-alive capacity.

Python script: The infeasible diagnostic example is not reliable or not shown.

Verification examples: Mixed-sense diagnostic examples are still missing.

Results: The reported large-case shortage pattern is plausible but not derived from the LP used in the code.

Results: Medium-case remote rover dispatch/delivered distinction is not shown.

Figures: Runtime and allocation figures are referenced before their values are fully consistent with the tables.

D) STOP

Review date: February 22, 2026

Audit request for: Final_Project_draft6.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Mission scenario, Data/assumptions, LP formulation, Implementation, Phase I explanation, Python script, Part 1 verification, Results, Runtime table

C) ISSUE NOTES (NO ADVICE):

Mission scenario: Source-backed mission framing is improved, but the connection between Foundation Surface Habitat and the chosen model is still slightly vague.

Data/assumptions: Source-status labels are better, but some load values are still labeled ambiguously as ranges rather than adopted values.

Data/assumptions: P_m and flexible capacity after keep-alive are still not consistently distinguished.

LP formulation: The flexible-load set F is defined, but h_0 still appears in places where it should be excluded from the optimization.

LP formulation: Nonnegativity is stated for wrong/incomplete variables.

LP formulation: Objective units are not clearly identified as weighted kWh-shortage.

Implementation: Tableau simplex and revised simplex are described, but shared preprocessing and basis initialization are not clearly stated.

Phase I explanation: Artificial-variable removal after Phase I is claimed but not supported by shown verification output.

Python script: The infeasible diagnostic problem is not clearly separated from the lunar benchmark.
Python script: The revised method does not clearly print its own objective/iteration count.
Part 1 verification: Expected output for the diagnostic LPs is missing or not checked against printed output.
Results: Large-case dormant slack is stated but not verified from dispatch/capacity values.
Runtime table: Some runtimes appear copied from earlier runs and do not match the current script output.
D) STOP

Audit request for: Final_Project_draft7.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Data/assumptions, LP formulation, Reduced crewed subproblem, Python script, Verification examples, Results, Runtime comparison

C) ISSUE NOTES (NO ADVICE):

Data/assumptions: Source-backed versus modeling-choice distinction is mostly present but still inconsistent for remote efficiencies and ISRU splits.

LP formulation: Final model statement does not fully match the standard-form matrix used by the script.

LP formulation: The distinction between source-side dispatch and delivered power is still easy to miss.

LP formulation: h_0 preprocessing is not stated clearly enough before defining the flexible model.

Reduced crewed subproblem: The simplification eliminates delivered variables but does not clearly state it is not the exact benchmark matrix.

Reduced crewed subproblem: The coefficient-ranking used to explain the large-case allocation is not fully justified.

Python script: The large-case capacity slack values are not printed in a way that directly verifies the report's interpretation.

Python script: Diagnostic outputs still use inconsistent labels compared with the report.

Verification examples: Infeasible case is mentioned but not reported with Phase I status.

Results: Large-case ISRU allocation values are stated without showing the source-side-to-delivered conversion.

Runtime comparison: Dataset-size counts are present but still do not clearly separate original variables from standard-form variables.

D) STOP

Review date: March 8, 2026

Audit request for: Final_Project_draft8.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Incorrect

B) SECTIONS WITH ISSUES:

Source/data consistency, Objective calculation, LP formulation, Python script, Verification examples, Results table, Runtime comparison

C) ISSUE NOTES (NO ADVICE):

Source/data consistency: Remote efficiencies 0.95, 0.92, and 0.90 are still presented too strongly as source-derived rather than assumed.

Source/data consistency: Some source-table rows still imply exact NASA values for adopted subsystem demands.

Objective calculation: The paper reports a large-case objective but does not show that it equals the weighted kWh-shortage from the reported shortages.

LP formulation: Fixed habitat preprocessing is described, but P_m versus R_m notation is not yet clean.

LP formulation: Nonnegativity and demand constraints are split across prose and equations, making the final LP harder to audit.

Python script: PuLP and custom-simplex models use different variable structures, but the report does not explain why their variable counts differ.

Python script: General-form preprocessing is present but not clearly connected to the lunar benchmark.

Verification examples: The feasible examples are present, but the infeasible example output is incomplete.

Results table: Objective values are rounded inconsistently between abstract, body, and table.

Runtime comparison: Runtime interpretation still suggests stronger performance conclusions than the small dataset set supports.

D) STOP**Audit request for: Final_Project_draft9.pdf**

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Data/assumptions, LP formulation, Implementation, Verification examples, Results, Runtime comparison, Acknowledgment/tools

C) ISSUE NOTES (NO ADVICE):

Data/assumptions: The source-status distinction is mostly fixed, but some load values are still labeled ambiguously as ranges rather than adopted model values.

Data/assumptions: Penalty weights are labeled as modeling choices, but the priority order is not fully explained.

LP formulation: Fixed habitat preprocessing is mostly correct, but P_m and R_m are still not emphasized consistently.

LP formulation: The reduced crewed subproblem can still be mistaken for the full LP benchmark.

Implementation: The general-form diagnostic examples are present, but the report does not yet connect them clearly to the lunar benchmark.

Implementation: Reduced-cost optimality convention for minimization is not stated clearly enough.

Verification examples: Phase I status for the infeasible example is not reported in the same style as the feasible examples.

Results: Solver objective and custom-simplex objective differ slightly, but the discrepancy is not explained.

Results: Delivered versus dispatched power is still under-explained for the remote rover and ISRU loads.

Runtime comparison: Runtime plot exists, but the dataset-size labels are still not fully clear.

Acknowledgment/tools: Tool-use statement is present but does not yet match the exact audit prompt being documented.

D) STOP**Audit request for: Final_Project_draft10.pdf**

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

LP formulation, Standard-form dimensions, Implementation, Results, Runtime comparison, Appendix

C) ISSUE NOTES (NO ADVICE):

LP formulation: Fixed habitat preprocessing is described, but the distinction between P_m and $R_m = P_m - 2$ is not emphasized consistently.

LP formulation: The final model is correct in substance, but constraints are split across prose and displayed equations.

Standard-form dimensions: The formula for simplex variables and constraints is not tied explicitly to n flexible loads.

Implementation: Phase I is described correctly, but the lunar benchmark's obvious slack basis is not clearly explained.

Implementation: Tableau and revised simplex use the same preprocessing, but this is not stated directly.

Results: Large-case objective is mostly traceable, but the objective calculation from shortages is still not shown.

Runtime comparison: The report compares runtime, but still risks implying a strong performance conclusion from very small instances.

Appendix: Part 1 verification output is referenced but not fully reproduced.

D) STOP

Audit request for: Final_Project_draft11.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Mission scenario, Data table, LP formulation, Results, Figure captions, Appendix

C) ISSUE NOTES (NO ADVICE):

Mission scenario: The Foundation Surface Habitat / Foundational Exploration terminology is closer, but the source connection is still slightly compressed.

Data table: Dormant ISRU split is listed, but its status as a modeling split of the total keep-alive value is not fully explicit.

Data table: Remote efficiencies are now assumption-labeled in some places but not all.

LP formulation: h_0 is properly excluded from F , but the exclusion should be easier to see.

Results: The medium-case remote rover dispatch/delivered distinction is not explained, which can make 0.526 dispatch versus 0.5 delivered look inconsistent.

Results: Large-case dormant slack is stated but not checked against reported dispatch values.

Figure captions: The allocation figure caption does not yet state that delivered power is after efficiency losses.

Appendix: Verification output is included, but formatting is uneven.

D) STOP

Review date: March 15, 2026

Audit request for: Final_Project_draft12.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Implementation, Reduced crewed subproblem, Results table, Runtime comparison, Acknowledgmen-

t/tools

C) ISSUE NOTES (NO ADVICE):

Implementation: Bland-style tie-breaking is mentioned, but the reduced-cost optimality convention for minimization is not stated clearly enough.

Implementation: Phase I is supported, but the report still slightly implies Phase I is needed for every lunar dataset.

Reduced crewed subproblem: The coefficient order $h_1, r_1, r_2, s, i_r, i_c$ is correct, but the weighted-service explanation is too terse.

Reduced crewed subproblem: The remaining crewed source-side power calculation is not shown near the allocation conclusion.

Results table: Large-case objective values are close but reported with inconsistent precision across solver, tableau, and revised rows.

Runtime comparison: Solver startup overhead is not distinguished from custom-matrix runtime.

Acknowledgment/tools: The audit/tool-use wording is present but not final.

D) STOP

Audit request for: Final_Project_draft13.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not.

Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Python script, Part 1 verification table, Results, Runtime table, Appendix

C) ISSUE NOTES (NO ADVICE):

Python script: The code now contains general-form preprocessing and Phase I logic, but the report table does not yet match the latest printed pivot counts.

Python script: The printed labels for tableau/revised outputs are not perfectly aligned with the report's table labels.

Part 1 verification table: Feasible lower-bound and mixed-sense examples are correct, but the infeasible example's Phase I status is not reported in the same format.

Results: The large-case dormant slack is stated but not checked against the reported dispatch values.

Results: Objective agreements are close, but the tolerance/rounding explanation is still missing.

Runtime table: Iteration counts are summarized, but it is not clear whether paired counts refer to tableau/revised or Phase I/Phase II.

Appendix: The appendix includes verification output, but the lunar benchmark output is only summarized.

D) STOP

Review date: March 28, 2026

Audit request for: Final_Project_draft14.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Data/assumptions, LP model statement, Results interpretation, Runtime comparison

C) ISSUE NOTES (NO ADVICE):

Data/assumptions: Penalty weights are now clearly modeling choices, but the priority order should match the coefficient-based allocation explanation.

LP model statement: Nonnegativity is stated, but the constraints are split across prose and displayed equations in a way that makes the final LP harder to audit.

LP model statement: P_m and R_m notation is correct but still slightly under-explained.

Results interpretation: The claim that crater-side ISRU is curtailed follows from the coefficient order, but that order is not yet tied directly to delivery efficiencies.

Results interpretation: The statement about full simultaneous crewed ISRU should stay tied to the selected model assumptions.

Runtime comparison: The timing conclusion is mostly reasonable, but the small scale of the instances should be clearer.

D) STOP

Audit request for: Final_Project_draft15.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Runtime comparison, Model dimensions, Figures, Appendix, Acknowledgment/tools

C) ISSUE NOTES (NO ADVICE):

Runtime comparison: Solver startup overhead is not distinguished from custom-matrix runtime, so the comparison can be misread.

Model dimensions: General LP variables and standard-form variables are both listed, but the difference between them is not fully explained.

Figures: Runtime figure references dataset load counts, but small-case load/constraint count needs clearer alignment.

Results interpretation: Acceptable, but the caption should state delivered power rather than source-side dispatch.

Appendix: Part 1 output is included, but the text should clarify that the longer lunar output is

summarized.

Acknowledgment/tools: Tool-use section still has minor duplicated/awkward wording.

D) STOP

Audit request for: Final_Project_draft16.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Abstract, Implementation, Results, Appendix, Acknowledgment/tools

C) ISSUE NOTES (NO ADVICE):

Abstract: No longer overclaims as stated accurately, but without the same rounding style used in result tables.

Implementation: The lunar benchmark's immediate BFS is correct, but the text does not explicitly state why Phase I is unnecessary for that benchmark.

Implementation: The paper should say the same preprocessing feeds both tableau and revised simplex.

Results: Objective discrepancy between PuLP/CBC and custom methods is small, but the tolerance/rounding explanation is missing.

Appendix: Verification output is included, but Part 1 output is shown while longer lunar output is summarized elsewhere; this should be noted.

Acknowledgment/tools: The LLM acknowledgment has duplicated headings and one spelling issue.

D) STOP

Review date: April 4, 2026

Audit request for: Final_Project_draft17.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Source/data consistency, Results, Runtime table, Acknowledgment/tools

C) ISSUE NOTES (NO ADVICE):

Source/data consistency: The table now separates direct source support from assumptions, but the

citation/modeling-note column is still not fully consistent for every row.

Source/data consistency: One row still suggests exact source support for an adopted modeling value.

Results: The objective discrepancy between PuLP/CBC and custom methods is now small, but the tolerance/rounding explanation is still missing.

Results: Dormant ISRU being fully served is correct, but the report should explicitly compare dormant dispatch to flexible capacity.

Runtime table: Iteration counts are summarized, but it is not clear whether the pair 11/11 refers to tableau/revised pivots.

D) STOP

Audit request for: Final_Project_draft18.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not.

Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Reduced crewed subproblem, Results, Runtime table, Figures

C) ISSUE NOTES (NO ADVICE):

Reduced crewed subproblem: The remaining crewed power calculation is correct, but it should be shown before assigning ridge-side ISRU dispatch.

Reduced crewed subproblem: The simplified calculation is correct but should remain clearly labeled as an interpretation aid.

Results: The large-case allocation table is internally consistent, but the text should explicitly say dormant ISRU is fully served because total dormant dispatch is below 38.

Results: The large-case qualitative conclusion is valid for the selected model but should not be overstated as a general engineering impossibility.

Runtime table: Iteration counts are summarized, but the pair notation remains slightly unclear.

Figures: Figure 1 notes that the small dataset is cut off, which is acceptable but should match the table directly.

D) STOP

Review date: April 11, 2026

Audit request for: Final_Project_draft19.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not.

Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Implementation, Results, Figures, Acknowledgment/tools

C) ISSUE NOTES (NO ADVICE):

Implementation: The revised simplex formula $\bar{c}_j = c_j - c_B^T B^{-1} A_j$ is correct, but free-index notation does not yet use the same notation consistently as tableau simplex.

Implementation: The sentence “validates the algorithm” is grammatically incomplete.

Results: The large-case qualitative conclusion is correct, but the sentence “resources have become a little scarce” is informal.

Results: Remaining crewed source calculation is proved, but the substitute difference is not computed in final text.

Figure caption: Figure 1 note says the small dataset is cut off, which is acceptable but should match the table directly.

Acknowledgment/tools: The tool-use paragraphs are present but still need smoother wording.

D) STOP**Audit request for: Final_Project_draft20.pdf**

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not.

Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Data/source table, Results table, Appendix, Acknowledgment/tools

C) ISSUE NOTES (NO ADVICE):

Data/source table: Source-backed versus source-informed categories are now mostly correct, but the table should not imply exact NASA values for all subsystem entries.

Results table: Solver and simplex objective values are compatible, but the absolute difference is not compared in the text.

Results table: Rounding is mostly consistent, but the abstract and table still use slightly different precision.

Appendix: Verification output is correct, but the appendix statement should clarify that the full lunar output is summarized elsewhere.

Acknowledgment/tools: Final tools wording is awkward; condense to one space.

D) STOP

Review date: April 24 and April 28, 2026

Audit request for: Final_Project_draft21.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic

manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Introduction, Implementation, Results interpretation, Minor consistency

C) ISSUE NOTES (NO ADVICE):

Introduction: The connection between diagnostic simplex testing and the lunar benchmark is grammatically incomplete, but the content is covered.

Implementation: Phase I infeasibility detection is now supported, but the report should avoid implying the lunar benchmark itself needs Phase I.

Results interpretation: The conclusion that full simultaneous crewed ISRU is unsupported is too strong without source capacity and priority-weight context; it should say this is true under the selected scenario.

Minor consistency: Some phrasing alternates between “Foundation Surface Habitat,” “Foundation Surface Habitat,” and “Foundational Exploration-inspired,” though the model meaning is consistent.

D) STOP

Audit request for: Final_Project_draft22.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Mostly correct

B) SECTIONS WITH ISSUES:

Acknowledgment/tools, Minor consistency, Formatting

C) ISSUE NOTES (NO ADVICE):

Acknowledgment/tools: The wording is adequate but a little awkward, with two acknowledgment headings and a typo in “documents.”

Minor consistency: Some phrasing alternates between “Foundation Surface Habitat” and “Foundation Surface Habitat,” though the meaning is consistent.

Formatting: A few table and figure references are visually corrected, but the mathematical content is covered.

D) STOP

Audit request for: Final_Project_draft23.pdf

Audit the attached Math 464 capstone paper and associated Python script for internal consistency, LP formulation correctness, source/data consistency, and logical validity (definition use, algebraic manipulation, cancellation/inverse rules). Only detect whether each section is well covered or not. Output must be extremely short: an overall status (Correct / Mostly correct / Incorrect), which

sections have issues, and brief notes on why they got flagged.

A) OVERALL STATUS:

Correct

B) PROBLEMS WITH ISSUES:

None detected.

C) ISSUE NOTES (NO ADVICE):

None.

D) STOP